```
#include <SoftwareSerial.h>
#include <DFRobotDFPlayerMini.h>
#include <Wire.h>
#include <Adafruit_MLX90614.h>
#include <PulseSensorPlayground.h>

#include <Wire.h>
#include <LiquidCrystal_I2C.h>

// Set the LCD address to 0x27 or 0x3F for a 16 chars and 2 line display
LiquidCrystal_I2C lcd(0x27, 16, 2); // Change 0x27 to 0x3F if your LCD uses that address

SoftwareSerial mySoftwareSerial(10, 11); // RX, TX
DFRobotDFPlayerMini myDFPlayer;

const int PulseWire = A0;
const int LED13 = 13;
int Threshold = 550;

PulseSensorPlayground pulseSensor;
Adafruit_MLX90614 mlx = Adafruit_MLX90614();

int currentFile = 1; // Start with the first file

void setup() {
  Serial.begin(9600);

  lcd.init();
  lcd.backlight();
  lcd.setCursor(0, 0); // Set cursor to column 0, row 0

  // Initialize DFPlayer Mini
  mySoftwareSerial.begin(9600);
  if (!myDFPlayer.begin(mySoftwareSerial)) {
    Serial.println("Unable to begin DFPlayer Mini:");
    Serial.println("1. Please recheck the connection!");
    Serial.println("2. Please insert the SD card!");
    while (true);
  }
```

```
  Serial.println("DFPlayer Mini online.");

  // Set volume (0 to 30)
  myDFPlayer.volume(30);

  // Play the first MP3 file
  myDFPlayer.play(currentFile);

  // Initialize PulseSensor
  pulseSensor.analogInput(PulseWire);
  pulseSensor.blinkOnPulse(LED13);
  pulseSensor.setThreshold(Threshold);

  if (pulseSensor.begin()) {
    Serial.println("PulseSensor started successfully!");
  } else {
    Serial.println("Error initializing PulseSensor.");
    while (1);
  }

  // Initialize MLX90614 sensor
  if (!mlx.begin()) {
    Serial.println("Error connecting to MLX90614 sensor. Check connections.");
    while (1);
  } else {
    Serial.println("MLX90614 sensor initialized successfully.");
  }
}

 void loop() {
  int myBPM = pulseSensor.getBeatsPerMinute(); // Get BPM from PulseSensor
  double objectTempC = mlx.readObjectTempC(); // Get object temperature in Celsius



  // Logic for playing specific files based on BPM and temperature
  if (objectTempC >= 33 && objectTempC <= 37 && myBPM >= 50 && myBPM <= 90 &&
myBPM > 20) {
    lcd.print( "Suhu kamu ");
    lcd.print(objectTempC);
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    playFile(4);
    delay(6000);
    lcd.clear();
    lcd.print( "Denyut kamu ");
    lcd.print(myBPM);
    playFile(3);
    delay(6000);
    lcd.clear();
  } else if (objectTempC >= 33 && objectTempC <= 37 && myBPM < 50 && myBPM > 20) {
    lcd.print( "Suhu kamu ");
    lcd.print(objectTempC);
    playFile(4);
    delay(6000);
    lcd.clear();
    lcd.print( "Denyut kamu ");
    lcd.print(myBPM);
    playFile(2);
    delay(6000);
    lcd.clear();
  } else if (objectTempC >= 33 && objectTempC <= 37 && myBPM > 90) {
    lcd.print( "Suhu kamu ");
    lcd.print(objectTempC);
    playFile(4);
    delay(6000);
    lcd.clear();
    lcd.print( "Denyut kamu ");
    lcd.print(myBPM);
    playFile(2);
    delay(6000);
    lcd.clear();
  } else if (objectTempC > 37 && myBPM >= 50 && myBPM <= 90 && myBPM > 20) {
    lcd.print( "Suhu kamu ");
    lcd.print(objectTempC);
    playFile(5);
    delay(6000);
    lcd.clear();
    lcd.print( "Denyut kamu ");
    lcd.print(myBPM);
    playFile(3);
    delay(6000);
```

```
    lcd.clear();
  } else if (objectTempC > 37 && myBPM < 50 && myBPM > 20) {
    lcd.print( "Suhu kamu ");
    lcd.print(objectTempC);
    playFile(5);
    delay(6000);
    lcd.clear();
    lcd.print( "Denyut kamu ");
    lcd.print(myBPM);
    playFile(1);
    delay(6000);
    lcd.clear();
  } else if (objectTempC > 37 && myBPM > 90) {
    lcd.print( "Suhu kamu ");
    lcd.print(objectTempC);
    playFile(5);
    delay(6000);
    lcd.clear();
    lcd.print( "Denyut kamu ");
    lcd.print(myBPM);
    playFile(2);
    delay(6000);
    lcd.clear();
  } else if (objectTempC < 33 && myBPM >= 50 && myBPM <= 90 && myBPM > 20) {
    lcd.print( "Suhu kamu ");
    lcd.print(objectTempC);
    playFile(6);
    delay(6000);
    lcd.clear();
    lcd.print( "Denyut kamu ");
    lcd.print(myBPM);
    playFile(3);
    delay(6000);
    lcd.clear();
  } else if (objectTempC < 33 && myBPM < 50 && myBPM > 20) {
    lcd.print( "Suhu kamu ");
    lcd.print(objectTempC);
    playFile(6);
    delay(6000);
    lcd.clear();
```

```
    lcd.print( "Denyut kamu ");
    lcd.print(myBPM);
    playFile(1);
    delay(6000);
    lcd.clear();
  } else if (objectTempC < 33 && myBPM > 90 ) {
    lcd.print( "Suhu kamu ");
    lcd.print(objectTempC);
    playFile(6);
    delay(6000);
    lcd.clear();
    lcd.print( "Denyut kamu ");
    lcd.print(myBPM);
    playFile(2);
    delay(6000);
    lcd.clear();
  } else if (objectTempC > 15 && myBPM == 0) {
    lcd.print( "Silahkan cek ");
    playFile(8);
    delay(3000);
    lcd.clear();
    lcd.print( "Silahkan cek ");
    playFile(8);
    delay(3000);
    lcd.clear();
  }

  // Check if playback is finished and restart if necessary
  if (myDFPlayer.available()) {
    if (myDFPlayer.readType() == DFPlayerPlayFinished) {
      currentFile++; // Move to the next file
      if (currentFile > 6) {
        currentFile = 1; // Loop back to the first file
      }
      myDFPlayer.play(currentFile); // Play the next file
    }
  }

  delay(4000);
```

```
}
void playFile(int fileNumber) {
  myDFPlayer.play(fileNumber); // Play the specified file number
  Serial.print("Playing file ");
  Serial.println(fileNumber);
}
```